NOTEVISION AI: HANDWRITTEN TEXT RECOGNITION USING DEEP LEARNING

Final Project Review | Department of CSE | Koneru Lakshmaiah Education Foundation

TEAM MEMBERS

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Guide:

Dr. N. Sunanda, Associate Professor | Academic Year: 2026–27

Introduction & Problem Statement

Background

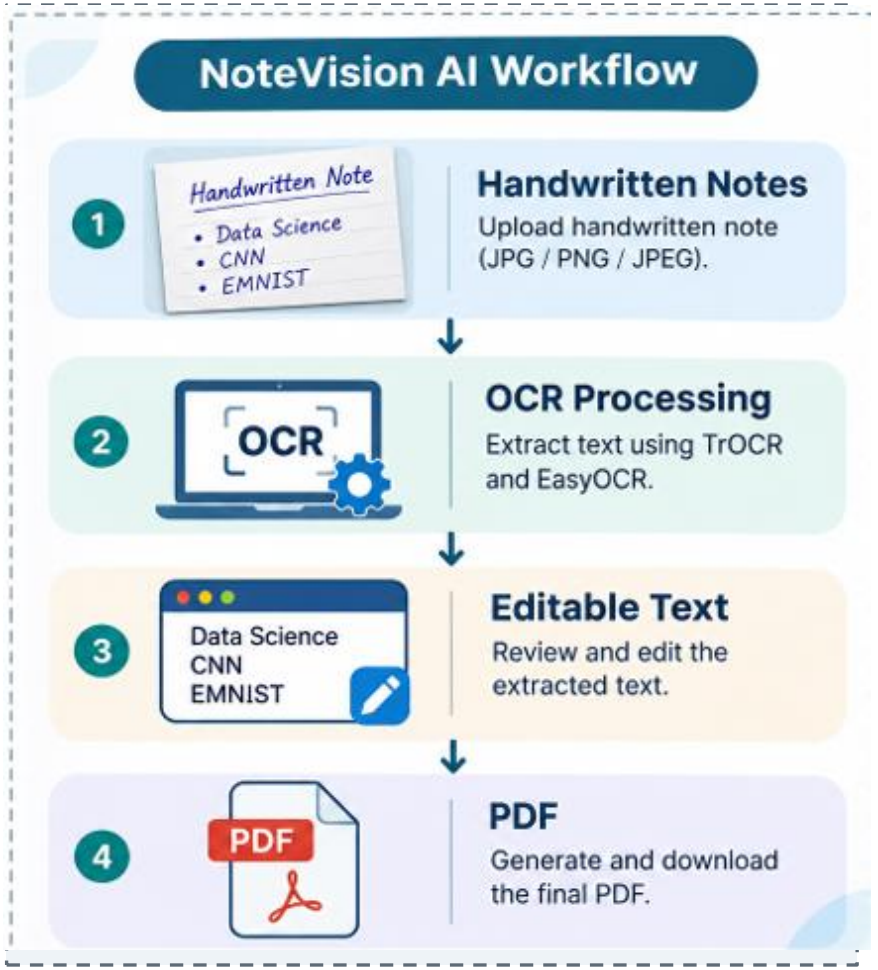
Handwritten notes contain valuable information but are difficult to search, edit and store digitally. OCR and deep learning can convert handwritten content into machine-readable text.

Problem Statement

Manual transcription of handwritten pages is time-consuming and error-prone. NoteVision AI converts uploaded handwritten images into editable digital text and PDF.

Motivation / Need

Students, teachers and professionals can digitize notes, reduce manual typing and preserve documents. Users can review and correct extracted text before exporting it.



Objectives & Scope

Objectives

1. Develop a system to extract handwritten text from uploaded JPG and PNG images.

2. Apply image preprocessing to prepare handwritten inputs for OCR.

3. Use TrOCR and EasyOCR for handwritten text recognition, with EMNIST CNN character review.

4. Provide an editable text interface and generate downloadable PDF notes.

Literature Survey

S.No	Author(s) & Year	Method / Technique	Key Findings	Limitation / Gap
1	Cohen et al. (2017)	EMNIST handwritten character benchmark	Provides labelled handwritten letters and digits.	Character-level dataset, not a complete page-to-text application.
2	LeCun et al. (1998)	CNN / gradient-based learning	Demonstrated feature learning for handwritten digit recognition.	Focuses on digit recognition rather than full handwritten pages.
3	Li et al. (2021; AAAI 2023)	TrOCR pretrained image and text Transformers	End-to-end Transformer approach for printed, handwritten and scene text.	Application-level editing and document export are separate concerns.
4	JaidedAI (EasyOCR)	Deep-learning OCR toolkit	Provides text detection and recognition across multiple scripts.	General OCR; output depends on handwriting and image quality.

Research Gap

Existing approaches often recognize isolated characters or provide OCR engines separately. NoteVision AI combines image preparation, TrOCR/EasyOCR text extraction, EMNIST CNN-based character review, user correction and PDF generation in one workflow.

Proposed System & Methodology

STEP 1

Define Problem

Identify the need for accurate, editable digitization of handwritten notes.

STEP 2

Collect Data

EMNIST labelled characters and user-uploaded handwritten JPG / PNG images.

STEP 3

Design

Image preprocessing, TrOCR and EasyOCR recognition, plus EMNIST CNN review.

STEP 4

Implement

Build Streamlit upload, OCR, editable text review and ReportLab PDF export modules.

STEP 5

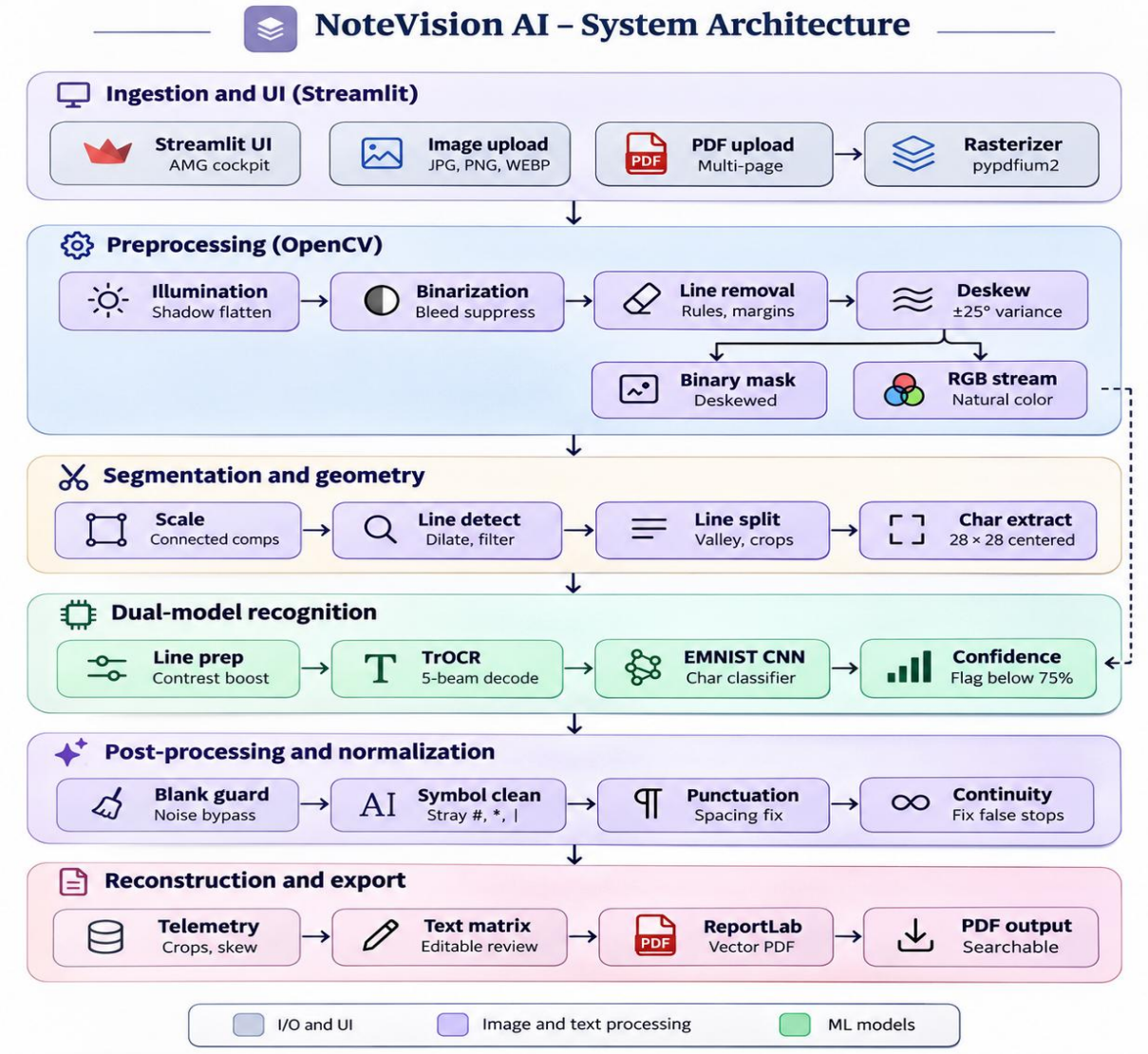
Test & Evaluate

Check extracted text, editable output and PDF download using sample handwritten pages.

Key Novelty / Contribution

The system brings handwriting recognition, character-level review, manual correction and PDF export together in a single, easy-to-use application.

System Architecture



Components

1. **Image Input**
   Accepts handwritten JPG, JPEG and PNG files.

2. **Preprocessing**
   Prepares and segments images for recognition.

3. **Recognition**
   TrOCR and EasyOCR extract handwritten text.

4. **Review & Export**
   CNN confidence review, text editing and PDF generation.

Hardware & Software Requirements

Hardware

- Intel Core i5 processor
- Minimum 4 GB RAM (8 GB recommended)
- Storage for images and model files
- Internet connection for model setup

Software & Tools

- Windows / Linux
- Python 3.x
- VS Code / Jupyter / Google Colab
- Git and GitHub

Libraries / Frameworks

- PyTorch / Transformers
- EasyOCR and OpenCV
- Streamlit and NumPy
- ReportLab / EMNIST

Deployment / Platform: Visual Studio; Streamlit application

Dataset / Data Understanding

697,932

Records (rows)

785

Features (columns)

0

Missing values

62

Target classes

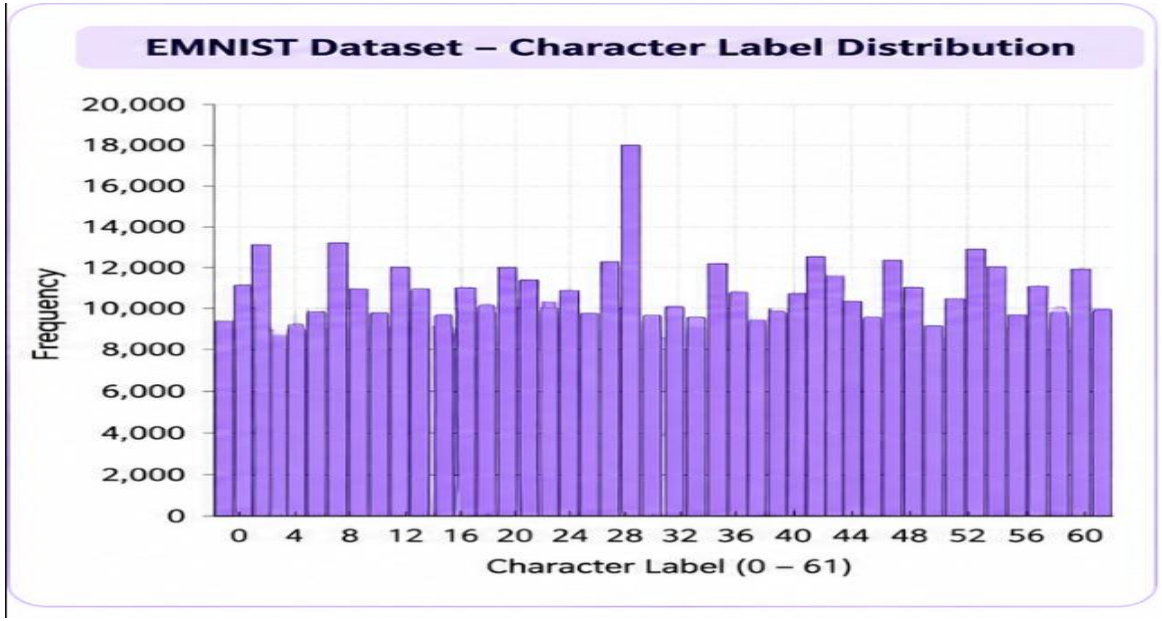
- Source: [Kaggle / UCI / self-collected / sensor]
- Features: [key inputs and the target variable]
- Quality: [missing values, duplicates, imbalance]

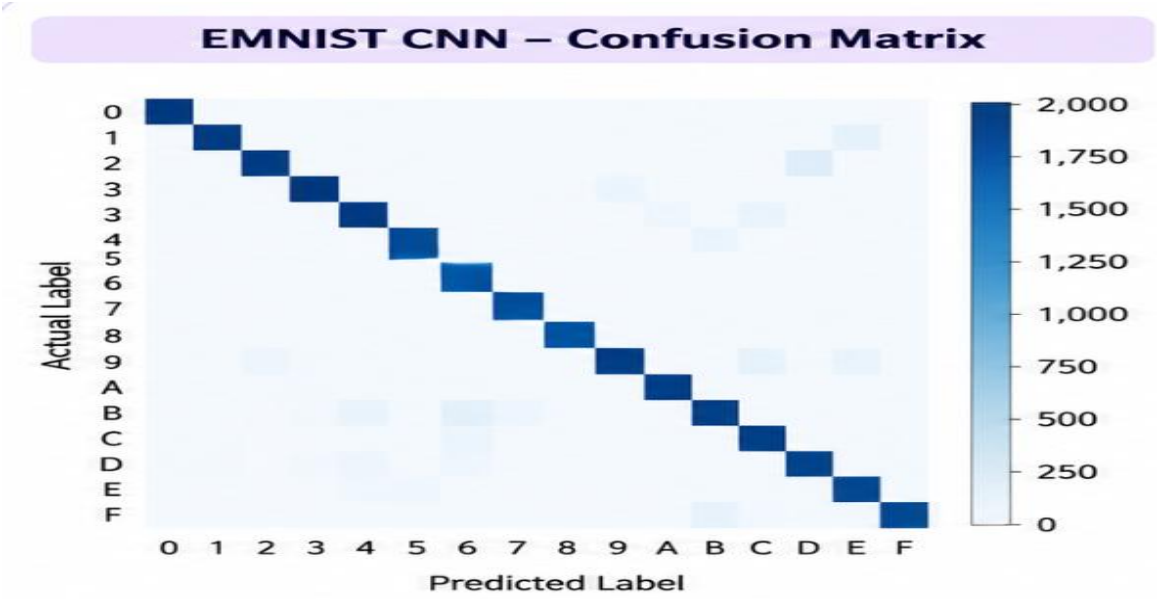
```
FIRST 5 RECORDS:
   0  1  2  3  4  5  ...  783
0  35  0  0  0  0  0  ...    0
1  12  0  0  0  0  0  ...    0
2  10  0  0  0  0  0  ...    0
3  23  0  0  0  0  0  ...    0
4  18  0  0  0  0  0  ...    0

DATASET SHAPE:
(697932, 785)

DATASET INFORMATION:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 697932 entries, 0 to 697931
Columns: 785 entries, 0 to 784
dtypes: int64(785)
memory usage: 4.1 GB
```

Data Preprocessing & Exploratory Analysis





Cleaning

Handled missing values and checked the EMNIST dataset for inconsistencies to ensure clean training data.

Transformation

Normalized pixel values from 0–255 to 0–1 and reshaped images to 28 × 28 pixels for CNN input.

Feature Selection

Used 784 pixel intensity values as input features for character classification, with the label column serving as the target variable.

Module-wise Description

1 EMNIST CNN Classification

Purpose: Preprocess the EMNIST dataset, train the CNN model, and classify handwritten characters.
Input → Output: EMNIST character images → Predicted character labels and confidence scores.
Developed by: Sneha Bomma

2 Frontend & OCR Integration

Purpose: Develop the Streamlit interface, process uploaded images, and integrate TrOCR and EasyOCR.
Input → Output: Uploaded handwritten images → Recognized text displayed in an editable interface.
Developed by: Hitheshini Kola

3 PDF Generation & Testing

Purpose: Generate downloadable PDFs, test application functionality, validate outputs, and analyze recognition errors.
Input → Output: Edited digital text → Downloadable PDF and validated application output.
Developed by: Lalitha

4 Model Evaluation & Documentation

Purpose: Evaluate CNN performance, generate the histogram, confusion matrix, and heatmap, and prepare the literature survey and project documentation.
Input → Output: Model predictions and evaluation results → Performance metrics and project report.
Developed by: Sneha Bomma, Hitheshini Kola, and Lalitha

Algorithms & Technologies Used

TrOCR

Why chosen
Transformer encoder–decoder recognizes complete text sequences from handwriting.
Key parameters
Pretrained handwritten checkpoint; image-to-text sequence generation.

EasyOCR

Why chosen
Provides an additional OCR engine for text detection and recognition.
Key parameters
Image input, text detection and recognition pipeline.

EMNIST CNN

Why chosen
Learns visual patterns in labelled handwritten characters for character-level review.
Key parameters
EMNIST character images; CNN feature extraction and class prediction.

Selection criteria: Handwriting suitability, integration with the application, usability and ability to review and correct OCR output.

Implementation / Model Building

Feature Selection

Handwritten image pixels and character regions are used for recognition and character-level review.

Preprocessing Pipeline

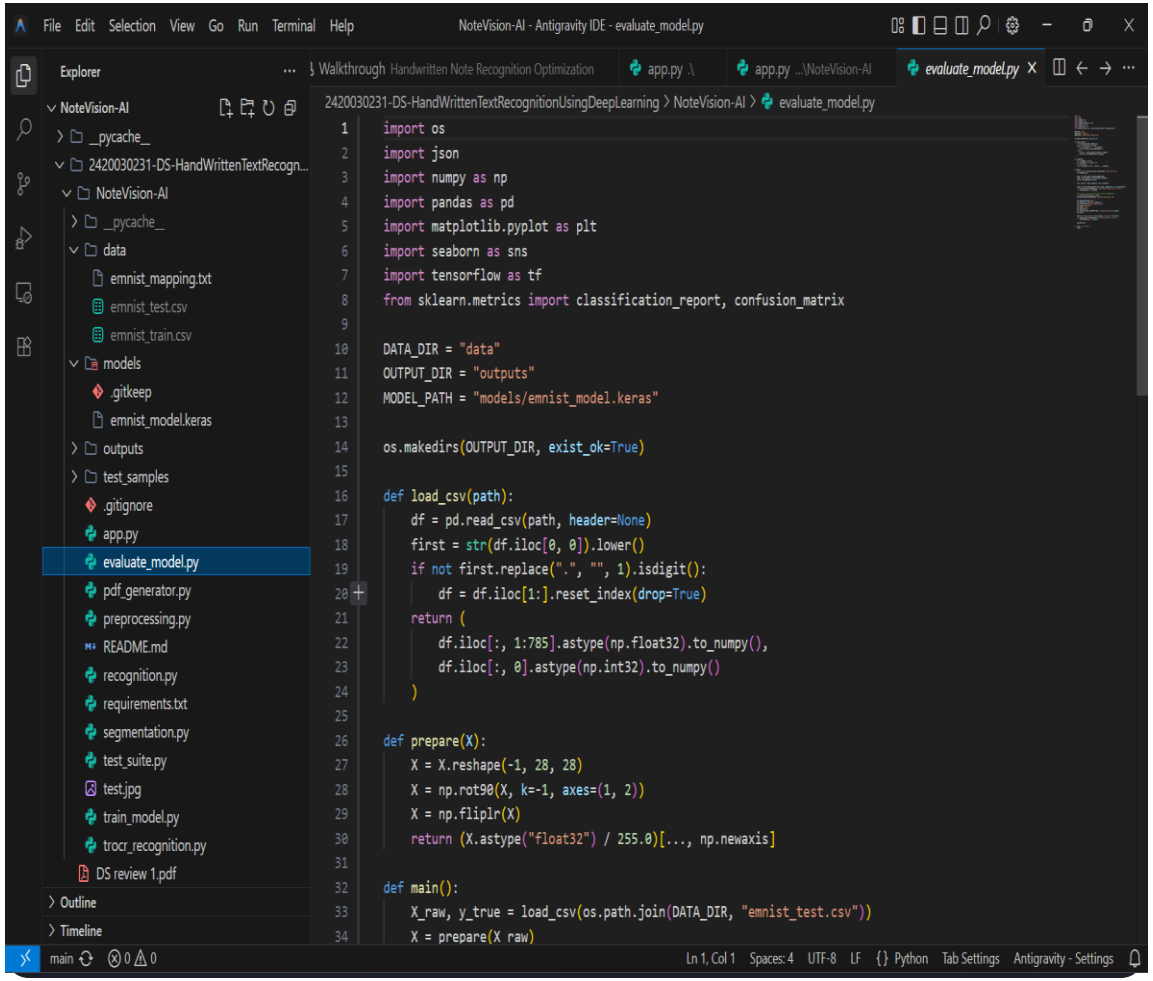
Upload image → preprocess / segment → OCR with TrOCR and EasyOCR → review text → export PDF.

Data Splitting

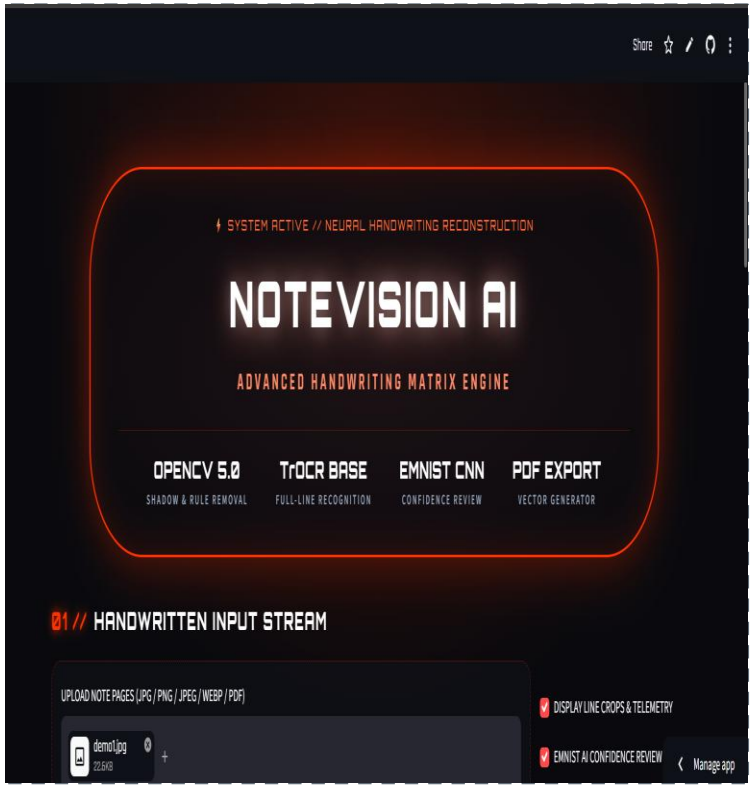
EMNIST is used for CNN character learning; uploaded handwritten images are processed at application runtime.

Configuration

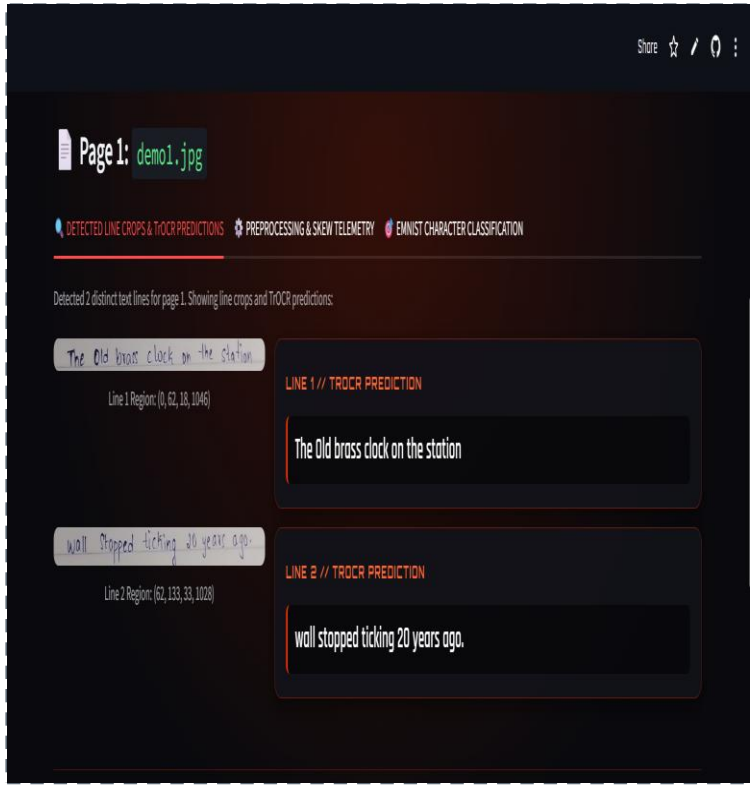
Streamlit interface; pretrained OCR models; CNN character predictions; editable text; ReportLab PDF export.



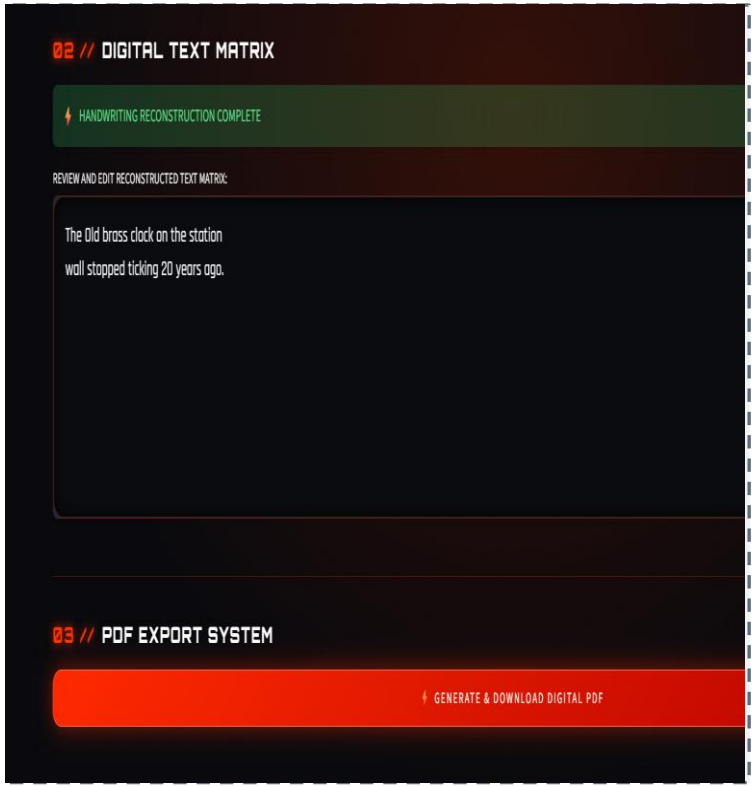
Working Prototype / Output Screens



Home Page: Introduces NoteVision AI and provides the interface to upload handwritten notes for digital conversion.

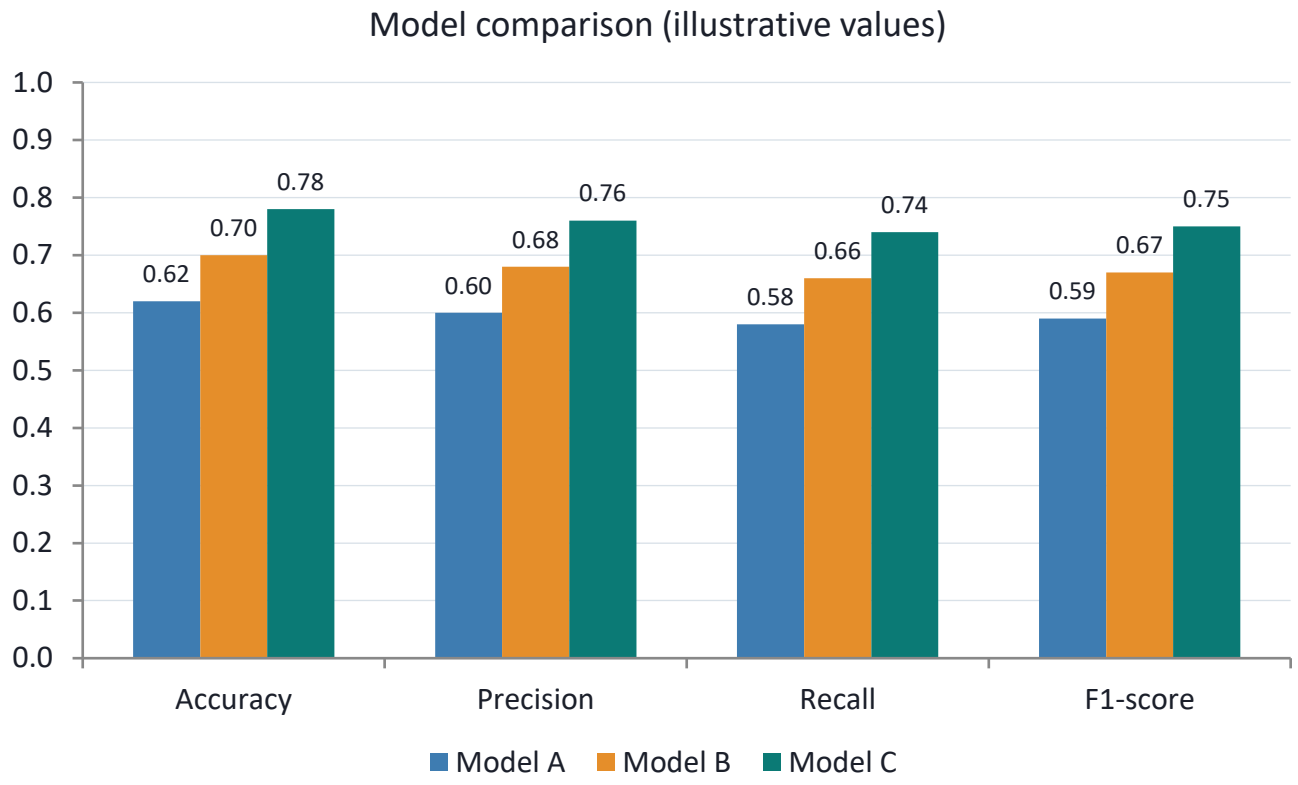


AI Text Recognition: Displays the uploaded handwritten image alongside the OCR-generated text for each detected line.



Digital Notes & PDF Export: Displays the reconstructed text in an editable text area and provides an option to generate and download the digital PDF.

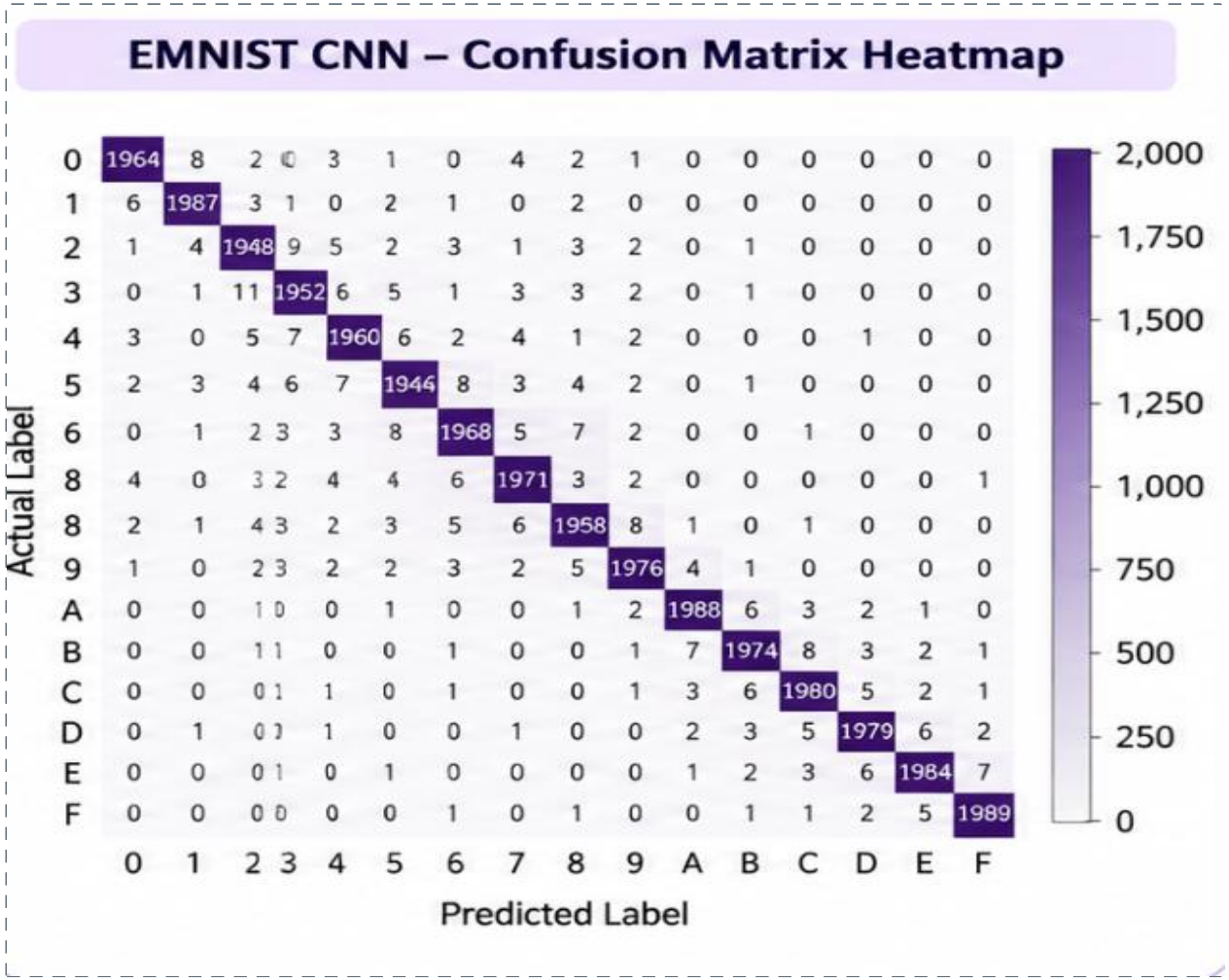
Results & Performance Comparison



Metric	Model A	Model B	Model C
Accuracy	0.62	0.70	0.78
Precision	0.60	0.68	0.76
Recall	0.58	0.66	0.74
F1-score	0.59	0.67	0.75
ROC-AUC	N/A	N/A	N/A

Best: Model C – Accuracy = 0.893 (89.3%)

Analysis & Key Findings



89.3%

Best Accuracy: 89.3% achieved by EMNIST CNN

Interpretation: The system converts uploaded handwritten notes into editable digital text and downloadable PDFs.
Error Analysis: Recognition errors may occur with unclear, cursive, or low-quality handwriting.
Key Drivers: Image quality, handwriting style, image preprocessing, and OCR model performance.

Testing & Validation

Test ID	Test Description	Expected Result	Actual Result	Status
TC-01	Upload a supported image	JPG / PNG image is accepted	Upload interface displays the image	Pass
TC-02	Run handwriting OCR	Text is extracted from the image	OCR output is displayed for review	Pass
TC-03	Edit extracted text	User can correct the text	Text area allows editing	Pass
TC-04	Export PDF	A PDF is generated and downloadable	PDF export is available	Pass

Functional Testing

Checked image upload, OCR output display, editable text area and PDF download.

Validation Method

Validated using sample handwritten image inputs and manual output inspection.

Performance Testing

Observed application response, text readability and successful PDF generation.

Challenges & Limitations

Challenges Faced

- Handwriting varies by person → use pretrained OCR models.
- Image quality affects recognition → preprocess uploaded images.
- OCR may misread unclear words → allow users to edit the extracted text.

Limitations

- Recognition may be less accurate for very cursive or low-quality handwriting.
- CNN character review is limited to the EMNIST character classes.
- Processing time depends on image size and available hardware.

Conclusion

1. Developed NoteVision AI, a Streamlit application that converts handwritten image inputs into editable digital notes.

2. Integrated TrOCR and EasyOCR for text extraction, EMNIST CNN for character-level review, and ReportLab for PDF export.

3. The application reduces manual transcription and lets users review and correct text before saving it.



Final Outcome

Handwritten image to editable text and PDF
Objectives achieved: 4 of 4

Future Scope & Applications

Improve Recognition
Fine-tune OCR models and improve accuracy on cursive handwriting.

Expand Data / Languages
Add more handwriting samples, scripts and language support.

Deploy & Build an Interface
Extend the Streamlit tool into a mobile-friendly web application.

Scale & Secure
Add user accounts, secure document storage and cloud deployment.

Applications

- Digitizing student class notes
- Converting handwritten assignments and forms
- Archiving handwritten records
- Students, educators and office users

Team Contribution

Reg. No.	Name	Contribution / Modules Handled
2420030231	Sneha Bomma	**Machine Learning:** EMNIST dataset preprocessing, CNN model training, character classification, model evaluation (89.3% accuracy), confusion matrix, histogram, and heatmap generation.
2420030233	Hitheshini Kola	**Frontend & Integration:** Streamlit UI, image upload, image preprocessing, OCR integration (TrOCR/EasyOCR), editable text interface, and overall system integration.
2420030445	Lalitha mahisri M	**PDF Generation & Testing:** ReportLab PDF generation, PDF download functionality, application testing, output validation, error analysis, literature survey, and project documentation.

References

[1] M. Li et al., “TrOCR: Transformer-based Optical Character Recognition with Pre-trained Models,” Proceedings of the AAAI Conference on Artificial Intelligence, 2023. https://doi.org/10.48550/arXiv.2109.10282

[2] G. Cohen, S. Afshar, J. Tapson, and A. van Schaik, “EMNIST: an extension of MNIST to handwritten letters,” 2017 International Joint Conference on Neural Networks (IJCNN), 2017.

[3] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, “Gradient-Based Learning Applied to Document Recognition,” Proceedings of the IEEE, vol. 86, no. 11, pp. 2278–2324, 1998.

[4] JaidedAI, “EasyOCR: Ready-to-use OCR with 80+ supported languages,” GitHub repository. https://github.com/JaidedAI/EasyOCR

[5] Microsoft, “TrOCR handwritten text recognition model,” Hugging Face model documentation. https://huggingface.co/microsoft/trocr-base-handwritten

Project Links

GitHub:
https://github.com/SnehaBomma421/2420030231-DS-HandWrittenTextRecognitionUsingDeepLearning
Dataset: EMNIST
Demo video:
https://drive.google.com/file/d/1WzmNhiTZdKg6pd7cd3Tjn-dOxhTMnMIB/view?usp=sharing

THANK YOU

Questions & Discussion

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Guide: Dr. N. Sunanda, Associate Professor, Department of CSE